

Zenghuang Fu

MSc Student, Institute of Automation, Chinese Academy of Sciences

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Summary

- Master's student at the Institute of Automation, Chinese Academy of Sciences, advised by Weiliang Meng and Xiaopeng Zhang. My research centres on multimodal large models and agentic reinforcement learning, with a focus on long-horizon planning and memory; I also work on video understanding, MLLM reasoning, object detection and multi-object tracking. I am actively looking for PhD positions.

Education

MSc Institute of Automation, Chinese Academy of Sciences , Pattern Recognition and Intelligent Systems • Advised by Weiliang Meng and Xiaopeng Zhang .	Beijing, China Sept 2024 – June 2027
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Publications

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| Robust detection in complex construction sites: HiPA-DETR with weather-aware and cross-domain generalization
Zenghuang Fu, Jinming Yang, Xiaofeng Han, Zhe Feng, Muyang Zhang, Changwei Wang, Weiliang Meng, Jiguang Zhang, Xiaopeng Zhang
(The Visual Computer) | May 2026 |
| Cognitive Scaffold: From Fluid Context to Crystallized Memory for Long-Horizon Deep-Research Agents
Qiuyuan Ai, Zenghuang Fu, Zhaoyang Li, Ping Jiang, Haoyu Wu, Jie Song, Guannan He
(ACL (Main)) | July 2026 |
| WM-DETR: Dual-Branch Wavelet-Mamba and Sparse Attention for Robust Underwater Object Detection
Xiaofeng Han, Shunpeng Chen, Zenghuang Fu, Longzhao Huang, Shengpeng Xu, Jinming Yang, Yixian Kong, Changwei Wang, Weiliang Meng, Xiaopeng Zhang
(Expert Systems with Applications) | June 2026 |
| Explicit temporal-semantic modeling for dense video captioning via context-aware cross-modal interaction
Mingda Jia, Weiliang Meng, Zenghuang Fu, Yiheng Li, Qi Zeng, Yifan Zhang, Ju Xin, Rongtao Xu, Jiguang Zhang, Xiaopeng Zhang
(AAAI) | Mar 2026 |
| HMR-1: Hierarchical Message Robot with Vision-Language-Model for Embodied Health-care
Rongtao Xu, Mingming Yu, Xiaofeng Han, Yu Zhang, Kaiyi Hu, Zhe Feng, Zenghuang Fu, Changwei Wang, Weiliang Meng, Xiaopeng Zhang
arxiv.org/abs/2603.08817 (arXiv preprint arXiv:2603.08817) | Mar 2026 |
| DMTrack: Deformable State-Space Modeling for UAV Multi-Object Tracking with Kalman Fusion and Uncertainty-Aware Association
Zenghuang Fu, Xiaofeng Han, Mingda Jia, Qi Zeng, Muyang Zhang, Changwei Wang, Weiliang Meng, Xiaopeng Zhang
arxiv.org/abs/2510.17860 (arXiv preprint arXiv:2510.17860) | Oct 2025 |

Multimodal fusion and vision-language models: A survey for robot vision

Aug 2025

Xiaofeng Han, Shunpeng Chen, Zenghuang Fu, Zhe Feng, Lue Fan, Dong An, Changwei Wang,
Li Guo, Weiliang Meng, Xiaopeng Zhang, Rongtao Xu, Shibiao Xu
(Information Fusion)

Projects

Awesome-Long-Horizon-Credit-Assignment-for-Agentic-RL

A curated reading list of papers, methods and resources on long-horizon credit assignment for agentic RL.

- <https://github.com/Zenghuang-Fu/Awesome-Long-Horizon-Credit-Assignment-for-Agentic-RL>

DMTrack

Reference implementation of the DMTrack UAV multi-object tracker.

- <https://github.com/Zenghuang-Fu/DMTrack>

Research Interests

Multimodal: Multimodal large models, MLLM reasoning, vision-language models

Agents: Agentic reinforcement learning, long-horizon credit assignment, memory and planning

Perception: Video understanding, object detection, multi-object tracking

Technical Skills

Programming: Python, PyTorch, C++

Language Skills

Chinese: Native speaker

English: Professional working proficiency